

LINEAR ALGEBRA – HOMEWORK 1

14 September, 2022

Due: 22 September, 2022

Textbook Problems. These problems will not be graded, but you must submit solutions to receive full credit for the homework.

Problem 1.1.3. If $\mathbf{v} + \mathbf{w} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $\mathbf{v} - \mathbf{w} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, compute and draw the vectors $\mathbf{v}$ and $\mathbf{w}$.

Problem 1.1.6. Every linear combination of

$$\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

has components that add up to what number? Find c and d so that $c\mathbf{v} + d\mathbf{w} = (3, 3, -6)$. Why is $(3, 3, 6)$ impossible?

Problem 1.1.9. If three corners of a parallelogram are $(1, 1)$, $(4, 2)$, and $(1, 3)$, what are all three of the possible fourth corners? Draw two of them.

Problem 1.1.25. Draw vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ so that their combinations $c\mathbf{u} + d\mathbf{v} + e\mathbf{w}$ fill only a line. Find vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ (in 3-dimensional space) so that their combinations $c\mathbf{u} + d\mathbf{v} + e\mathbf{w}$ fill only a plane.

Problem 1.1.26. What combination of $c \begin{bmatrix} 1 \\ 2 \end{bmatrix} + d \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ produces $\begin{bmatrix} 14 \\ 8 \end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

Problem 1.1.29. Find two different combinations of the three vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$, and $\mathbf{w} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ that produce $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. If you take any three vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in the plane, will there always be two different combinations that produce $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$?

Graded Problem.

Consider the three vectors: $\mathbf{u} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$.

- Describe geometrically (line, plane, or all of 3-dimensional space) the set of all linear combinations of $\mathbf{u}$ and $\mathbf{v}$. Also, find the components of a general linear combination of $\mathbf{u}$ and $\mathbf{v}$.
- Describe geometrically the set of all linear combinations of $\mathbf{v}$ and $\mathbf{w}$. Also, find the components of a general linear combination of $\mathbf{v}$ and $\mathbf{w}$.
- Which vectors in 3-dimensional space are linear combinations of $\mathbf{u}$ and $\mathbf{v}$ and *also* are linear combinations of $\mathbf{v}$ and $\mathbf{w}$? (In other words, the intersection of two distinct _____ in 3-dimensional space is a _____.)